
0.1 My (?) theorem

Rigorous revision: status, conventions, and conditional result

This subsection replaces the exploratory calculation above. The earlier text is retained in the source inside an `\iffalse` block so that the provenance of the formulas remains visible. Line numbers in the audit below refer to the pre-revision file.

What is and is not established

The stable-map calculation of Faber–Pandharipande is a theorem on $\overline{M}_{g,0}(\mathbb{P}^1, d)$: it uses its standard equivariant perfect obstruction theory and virtual localization. The available pseudo-stable sources establish facts about $\overline{M}_{g,n}^{\text{ps}}$, its Hodge classes, and the contraction of elliptic tails. They do not, in the files available here, construct a proper Deligne–Mumford stack $\overline{\mathcal{M}}_{g,0}^{\text{ps}}(\mathbb{P}^1, d)$ with an equivariant perfect obstruction theory and prove a virtual localization theorem for it. The file `DustyNotes .pdf` proposes the required graph rules and tests them in examples; they do not supply that missing construction. Accordingly, the result below is conditional.

Fix the following conventions. The tangent weights of $\mathbb{P}^1$ at 0 and ∞ are respectively

$$w_0 = t, \quad w_\infty = -t.$$

The two copies of $\mathcal{O}_{\mathbb{P}^1}(-1)$ have fiber weights $[0, t]$ and $[-t, 0]$. Thus, if $z = x^d$ on the degree- d edge, the tangent weights on the source at its two ends are t/d and $-t/d$. We use the homogeneous Hodge polynomial

$$\Lambda_g^{\text{ps}}(t) := \sum_{i=0}^g t^{g-i} \lambda_i^{\text{ps}}.$$

Here $\mathbb{E}^{\text{ps}} := \pi_* \omega_\pi$ on the moduli stack of pseudo-stable curves and $\lambda_i^{\text{ps}} := c_i(\mathbb{E}^{\text{ps}})$. This convention must also be the convention used by any future obstruction theory on the pseudo-stable map space. In particular,

$$e(\mathbb{E}^\vee \otimes \mathbb{C}_t) = (-1)^g \Lambda_g^{\text{ps}}(-t), \quad e(\mathbb{E}^\vee \otimes \mathbb{C}_{-t}) = (-1)^g \Lambda_g^{\text{ps}}(t).$$

We isolate the assumptions that are not supplied by the available sources.

- (A1) There is a proper pseudo-stable map stack with a $\mathbb{C}^*$ -equivariant perfect obstruction theory whose restriction to the nodal one-edge loci is represented by the usual complex $R\pi_* f^* T\mathbb{P}^1$, together with the usual deformation theory of the source, and virtual localization holds for this theory.
- (A2) For the chosen linearizations, the only nonzero graph contributions are the proposed one-edge graphs. A genus-one decoration

at an end is represented by the rational-cusp vertex of `DustyNotes.pdf`, not by a contracted elliptic tail.

(A3) For $g \geq 2$ and $0 \leq k \leq g-1$, the following integrated comparison identity holds:

$$\begin{aligned} \int_{\overline{M}_{g,1}^{\text{ps}}} \lambda_g^{\text{ps}} g_*^1 \left(\prod_{j=1}^k (\psi_{\star j} - \psi_{\bullet j}) \right) \psi_P^{2g-2k-2} \\ = \left(\int_{\overline{M}_{g-k,1+k}} \lambda_{g-k} \psi_P^{2g-2k-2} \prod_{j=1}^k \psi_{\star j} \right) \left(\int_{\overline{M}_{1,1}} \lambda_1 \right)^k. \end{aligned}$$

The formulas cited in the old draft for linear pseudo-stable Hodge integrals and for the quadratic pseudo-stable Mumford class do not, by themselves, prove this product identity. It requires a projection-formula and self-intersection calculation for the elliptic-tail strata.

Theorem 0.1.1 (Conditional one-edge factorization). *Assume (A1)–(A3) and let $g_1, g_2 \geq 2$. Let F_{g_1, g_2} be the one-edge fixed locus consisting of a degree- d rational component, a contracted genus- g_1 component over 0, and a contracted genus- g_2 component over ∞ . Then*

$$F_{g_1, g_2} \cong [(\overline{M}_{g_1,1}^{\text{ps}} \times \overline{M}_{g_2,1}^{\text{ps}}) / \mu_d].$$

After removing the residual factor $1/d$ coming from μ_d , its localization contribution is

$$I_{(g_1, g_2)} = d^2 I_{g_1} I_{g_2},$$

where, for $g \geq 2$,

$$I_g := \int_{\overline{M}_{g,1}^{\text{ps}}} \frac{\lambda_g^{\text{ps}} \Lambda_g^{\text{ps}}(t) (-1)^g \Lambda_g^{\text{ps}}(-t)}{t^2 (1 - d\psi_P/t)}. \quad (0.1.1)$$

The full stack contribution of this locus is therefore $d I_{g_1} I_{g_2}$.

Proof. The deformation–obstruction sequence, restricted to moving parts, is

$$0 \rightarrow \text{Aut}(C) \rightarrow H^0(C, f^* T\mathbb{P}^1) \rightarrow T_{[f]}^1 \rightarrow \text{Def}(C) \rightarrow H^1(C, f^* T\mathbb{P}^1) \rightarrow T_{[f]}^2 \rightarrow 0$$

Here $\text{Def}(f) := H^0(C, f^* T\mathbb{P}^1)$ and $\text{Ob}(f) := H^1(C, f^* T\mathbb{P}^1)$. Consequently the convention used in the old draft is

$$e(N^{\text{vir}}) = \frac{e(\text{Def}(f))^{\text{mov}} e(\text{Def}(C))^{\text{mov}}}{e(\text{Aut}(C))^{\text{mov}} e(\text{Ob}(f))^{\text{mov}}}.$$

The degree- d component has a zero-weight infinitesimal scaling; it is fixed and hence does not enter this Euler class. Its residual finite automorphism group is μ_d , which produces the separate stack factor $1/d$.

The moving map deformations on the edge have weights $t - kt/d$, for $0 \leq k \leq 2d$, with the zero weight $k = d$ removed. Hence

$$e(H^0(\mathbb{P}^1, f^* T\mathbb{P}^1)^{\text{mov}}) = (-1)^d \left(\frac{t}{d} \right)^{2d} (d!)^2.$$

The two obstruction-bundle insertions restricted to the edge contribute

$$(-1)^{d-1} \left(\frac{t}{d} \right)^{2d-2} ((d-1)!)^2.$$

Their quotient is exactly

$$\frac{(-1)^{d-1} (t/d)^{2d-2} ((d-1)!)^2}{(-1)^d (t/d)^{2d} (d!)^2} = -\frac{1}{t^2}.$$

Let P_i be the attaching marking on the contracted component. The two node-smoothing weights are

$$\frac{t}{d} - \psi_{P_1} = \frac{t}{d} \left(1 - \frac{d\psi_{P_1}}{t} \right), \quad -\frac{t}{d} - \psi_{P_2} = -\frac{t}{d} \left(1 + \frac{d\psi_{P_2}}{t} \right).$$

After inversion their product contributes

$$-\frac{d^2}{t^2} \frac{1}{(1 - d\psi_{P_1}/t)(1 + d\psi_{P_2}/t)}.$$

Combining this with the edge quotient gives d^2/t^4 . The two local-curve obstruction insertions and the two factors from $H^1(C, f^*T\mathbb{P}^1)$ supply the Hodge numerators, so the remaining class is

$$d^2 \left(\frac{\lambda_{g_1}^{\text{ps}} \Lambda_{g_1}^{\text{ps}}(t) (-1)^{g_1} \Lambda_{g_1}^{\text{ps}}(-t)}{t^2 (1 - d\psi_{P_1}/t)} \right) \cdot \left(\frac{\lambda_{g_2}^{\text{ps}} \Lambda_{g_2}^{\text{ps}}(-t) (-1)^{g_2} \Lambda_{g_2}^{\text{ps}}(t)}{t^2 (1 + d\psi_{P_2}/t)} \right).$$

This proves the claimed factorization before the μ_d -quotient.

It remains to compare the two tail integrals. The homogeneous quadratic relation has terms of codimension $2k$ multiplied by t^{2g-2k} . Since $\dim \overline{M}_{g,1}^{\text{ps}} = 3g - 2$, the only term from the geometric series that can pair with it and λ_g^{ps} is $\psi_P^{2g-2k-2}$. This exponent is even. Replacing t by $-t$ and $d\psi_P/t$ by $-d\psi_P/t$ therefore does not change the integral. Moreover, the factor t^{2g-2k} , the prefactor t^{-2} , and the chosen geometric-series term cancel to total equivariant degree zero. This is the precise reason that the two tails agree; it is not a general permission to discard the sign of ψ . $\square$

Evaluation of the tail and the exceptional genera

Set

$$b_0 := 1, \quad b_g := \int_{\overline{M}_{g,1}} \psi_1^{2g-2} \lambda_g = \frac{2^{2g-1} - 1}{2^{2g-1}} \frac{|B_{2g}|}{(2g)!} \quad (g \geq 1).$$

Under assumption (A3), the λ_g -formula gives, for $g \geq 2$,

$$I_g = \sum_{k=0}^{g-1} \frac{d^{2g-2k-2}}{k!} \frac{(2g-k-2)!}{(2g-2k-2)!} \frac{b_{g-k}}{24^k}. \quad (0.1.2)$$

Indeed,

$$\int_{\overline{M}_{g-k, 1+k}} \lambda_{g-k} \psi_P^{2g-2k-2} \prod_{j=1}^k \psi_{\star j} = \frac{(2g-k-2)!}{(2g-2k-2)!} b_{g-k},$$

and $\int_{\overline{M}_{1,1}} \lambda_1 = 1/24$. The upper bound is $g-1$, because the exponent of ψ_P must be nonnegative.

There is no geometric integral (0.1.1) for an unstable genus-zero vertex, and a genus-one end of a pseudo-stable map is a rational cusp rather than a contracted elliptic tail. Thus the values

$$I_0 := \frac{1}{d^2}, \quad I_1 := \frac{d^2-1}{24d^2}$$

are conventions encoding, respectively, the unstable vertex and the genus-one cusp rule proposed in [DustyNotes .pdf](#). They are not obtained from negative factorials and they remain conditional on (A1)–(A2). With these conventions the uniform bookkeeping identity

$$I_{(g_1, g_2)} = d^2 I_{g_1} I_{g_2}$$

also covers ends decorated by 0 or 1.

Generating functions and coefficient conventions

As a formal power series, let

$$B(u) := \sum_{g \geq 0} b_g u^{2g} = \frac{u/2}{\sin(u/2)}, \quad \mathcal{J}(u) := \sum_{g \geq 0} I_g u^{2g}.$$

Then

$$\mathcal{J}(u) = \frac{1}{d^2} \left(1 - \frac{u^2}{24}\right) B\left(\frac{du}{1 - u^2/24}\right) = \frac{u/(2d)}{\sin\left(\frac{du}{2(1 - u^2/24)}\right)}. \quad (0.1.3)$$

This statement is purely formal; no appeal to analytic Fubini is needed. Equivalently,

$$I_g = \sum_{n=0}^g d^{2n-2} b_n [x^{g-n}] (1 - x/24)^{1-2n}.$$

For $n = 0$, the factor is $1 - x/24$; this gives I_0 and the correction $-1/(24d^2)$ in I_1 , without assigning values to negative factorials. For $g \geq 2$, coefficient extraction is exactly (0.1.2). Conversely, for $g \geq 2$,

$$b_g = d^{2-2g} \left(I_g - \sum_{k=1}^{g-1} \frac{d^{2g-2k-2}}{k!} \frac{(2g-k-2)!}{(2g-2k-2)!} \frac{b_{g-k}}{24^k} \right).$$

If the proposed graph sum in (A1)–(A2) is valid for every genus, then the full invariant has the conditional generating series

$$\sum_{g \geq 0} C^{\text{ps}}(g, d) u^{2g} = d \mathcal{J}(u)^2 = \frac{u^2/(4d)}{\sin^2\left(\frac{du}{2(1 - u^2/24)}\right)}. \quad (0.1.4)$$

By comparison, the established stable-curve calculation of Faber-Pandharipande gives

$$C(g, d) = d^{2g-3} \sum_{g_1+g_2=g} b_{g_1} b_{g_2}, \quad \sum_{g \geq 0} C(g, d) u^{2g} = d^{-3} B(du)^2.$$

What transfers to the pseudo-stable calculation is the edge weight algebra, the two smoothing factors on a nodal one-edge locus, the residual μ_d -factor, and the stable λ_g -integral used after a valid comparison to $\overline{M}_{g,n}$. What does not transfer without proof is the map stack, its virtual class, localization, the completeness of the fixed graph list in the presence of cusps, the cusp-vertex deformation factors, and the integrated identity (A3).

Audit of the original argument

The classifications below refer to the original line numbers.

- (a) **My()Theorem.tex, lines 29–31; DoctoralNotebookJIRR.tex, lines 1554–1556 — conjectural.** The formula treats a pseudo-stable virtual class and localization as available. Neither `DustyNotes .pdf` nor `SchubertNewComp .tex` proves the existence of the required map-space obstruction theory. Replacement: assumptions (A1)–(A2) above.
- (b) **My()Theorem.tex, lines 33–37 — incorrect as a complete fixed-locus statement.** The listed nodal graphs omit rational cuspidal source components, which `DustyNotes .pdf`, pp. 1–5, explicitly identifies as new fixed loci. The list is usable only after the linearization vanishing and the cusp rules are proved.
- (c) **My()Theorem.tex, lines 82–124 — ambiguous.** Two incompatible definitions of Λ_g^{ps} occur. The homogeneous definition fixed above is the one required by the Euler classes.
- (d) **My()Theorem.tex, lines 141–206 and 876–906 — correct stable weight algebra, conjectural pseudo-stable application.** With the declared weights, the edge quotient is $-t^{-2}$, the smoothing quotient is $-d^2 t^{-2}$, and their product is $d^2 t^{-4}$. The same computation on a pseudo-stable map space depends on (A1).
- (e) **My()Theorem.tex, line 876 — ambiguous.** “There are no automorphisms” must mean that there is no moving infinitesimal automorphism. The edge has a fixed infinitesimal scaling and the map has residual automorphism group μ_d ; the latter gives $1/d$.
- (f) **My()Theorem.tex, lines 909–915 and 918–946 — incomplete.** Evenness of the surviving ψ -power follows from the dimension calculation above. The last displayed equality is not a class identity, as the old draft itself observes; the needed integrated version is precisely (A3).

- (g) **My0Theorem.tex, lines 951–986 — incorrect/incomplete.** The edge-coloring calculation is explicitly marked wrong and cannot prove (A_3) . It should be replaced by a genuine pullback, self-intersection, and excess-intersection computation for the gluing morphisms.
- (h) **My0Theorem.tex, lines 997–1005 — incomplete.** The coefficient is correct for $0 \leq k \leq g-1$, conditional on (A_3) . The printed upper limit g is wrong. Formula (0.1.2) is the corrected statement.
- (i) **My0Theorem.tex, lines 1007–1024 and 1035–1096 — incorrect as geometry, correct as later bookkeeping.** Genus zero is an unstable vertex and genus one is a cusp vertex. Neither is the integral (0.1.1). The negative-factorial manipulations at lines 1043–1062 are invalid. Formal coefficient extraction gives the same stated values rigorously.
- (j) **My0Theorem.tex, lines 1098–1162 — correct formal identity after correction of the derivation.** The closed generating function is correct, but the proof must use generalized binomial series, not analytic Fubini or negative factorials. See (0.1.3).
- (k) **My0Theorem.tex, lines 1186–1201 — conjectural.** The convolution is algebraically correct once all vertex values and the $1/d$ stack factor are accepted; its interpretation as a pseudo-stable Gromov–Witten invariant still requires (A_1) – (A_2) .
- (l) **My0Theorem.tex, lines 1225–1231 — incorrect.** The displayed differential equation for B fails already in degree zero. A correct elementary relation is

$$\frac{B'(u)}{B(u)} = \frac{1}{u} - \frac{1}{2} \cot(u/2).$$

- (m) **FaberPandharipande.tex, lines 1301–1357 — correct in the stable setting.** These lines prove the one-edge reduction and the formula for $C(g, d)$ only on the ordinary stable-map space. Lines 1383–1412 correctly give b_g and its generating series. They do not justify a pseudo-stable virtual localization formula.
- (n) **SchubertNewComp.tex, lines 622–675 — correct but insufficient for maps.** The construction concerns the moduli space of pseudo-stable curves. It supplies neither a pseudo-stable map stack nor a perfect obstruction theory or virtual localization theorem.